Long-Run Dynamics of Blockwise Self-Conditioned Language Generation After Prompt Eviction

Semantic Afterlife

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Abstract

What persists when a language model continues from its own output after the original prompt has left its input? We study blockwise re-prompting: each call receives the current finite tail, and stop events are logged without terminating the trajectory. We distinguish reaching the prescribed horizon, entering a textual repetition lock, and a representation-space gap between trajectories from different versus identical initial texts. A hosted occupancy replication establishes a positive last-band gap in all three representation spaces; BGE-M3 gives 0.208 with a 95% interval of $[0.091, 0.314]$. The local family matrix compares Base and post-trained checkpoints of Qwen, OLMo, Ministral, and Gemma. Completion ordering differs by family. Qwen and Ministral have positive NF4 gaps in both representation spaces, whereas Gemma Base has an established negative BGE-M3 gap and an unestablished Qwen-embedding sign. Qwen locks are usually confirmed before full prompt eviction and persist beyond it. In a separate extension of selected locks, 0/8 confirmed escapes are observed through 4,999,168 original tokens per trajectory, with censoring at 1220.5 window turnovers from the first token of the short-horizon series. These are conditional, finite-horizon observations of surface repetition and initial-text dependence. They establish neither recovered semantic memory nor validated metastable semantic states. The measured kernel uses fresh block prompts, rather than native sliding attention; its results do not directly establish the behaviour of ordinary chat or autonomous agents.

1 Introduction

Let W be an imposed sliding window in generator tokens and B the block requested at each call. The object of this paper is not token-by-token sliding attention. It is a block re-prompt of the model's own tail. At step k the input X_k is the last $\min(|\cdot|, W)$ tokens of the seed followed by everything generated so far. The model emits a block and the window keeps only the tail:

$$Y_k^{(B)} \sim P_\theta(\cdot \mid X_k; B, \text{temperature}), \qquad |Y_k^{(B)}| \le B, \qquad X_{k+1} = \text{Tail}_W\big(X_k \oplus Y_k^{(B)}\big). \tag{1}$$

Each call is fresh. Position indices restart, and no key–value cache is carried from the previous call. When the model emits a stop token, the stop is logged and the tail is sent again. The measured object is externally sustained self-continuation after repeated attempts to finish. It is not an unrestricted chain whose absorbing state is end-of-sequence, and it is not ordinary chat or an autonomous agent.

The horizon is the generated-token count at which the seed has fully left the window. For a non-empty seed that count is W, which is turnover $T/W = 1$, where T is the number of generated tokens. Eviction can start earlier, at $W - L_0$, when the window first fills and the next token can drop a seed token. Those two times are not the same. Full eviction is turnover 1.

The question that the measurements support is narrower than a theory of semantic memory. After the original prompt has been physically removed, can self-continuation preserve textual repetitions, and can trajectories remain dependent on which initial text they started from? Do completion and the stability of those regimes differ across families and checkpoints? The answer, on the processes actually run, is that repetitions can persist for a long time, that a gap between same-seed and different-seed trajectories can remain positive

after eviction, and that both completion and that gap depend on the family and the checkpoint. Section 4 presents these results. Section 4.5 records the questions the same programme did not establish, including a semantic-memory half-life, validated metastable states, and transition currents. These limits determine the scope of the conclusions.

2 Related work

Textual repetition under neural decoding is a known regime. Holtzman et al. (2020) show that likelihood-maximising decoding falls into repetitive loops, and propose nucleus sampling as a remedy inside a single prompt-conditioned generation. Xu et al. (2022) measure a self-reinforcing loop: a sentence that has already appeared in the context becomes more likely to be continued. Both papers analyse degeneration inside one generation. They are the right alternative explanation of a lock. They are not a measurement of a tail that has been fed back as the next prompt after the original text has left a finite window.

Zekri et al. (2024) represent an autoregressive model with a finite window as a Markov chain on token strings and prove a unique stationary distribution for their positive-temperature, full-support token-sampling kernel. Under its assumptions, that theorem rules out several stationary laws on the token chain. It does not measure occupancy after the seed has been evicted, and it does not license reading a positive embedding gap as a second stationary distribution.

Cycles also appear when a model is asked to transform a whole text. Wang et al. (2025) iterate paraphrasing and find low-order cycles, typically of period two, that survive increases in temperature. Geng et al. (2026) define a Markovian generation chain in which each step sees a prompt template plus the previous output and no earlier history. On iterative rephrasing and round-trip translation they already measure lexical and semantic drift between successive iterations, including with overlap scores and embedding similarity, and they distinguish early exact recurrence from longer transients. Their step replaces the whole text under an instruction. Equation (1) appends a block to a growing stream and then discards every token outside the last W, including the original prompt once turnover reaches 1. Telephone-game protocols likewise pass the whole previous text under an instruction (Perez et al., 2025; Mohamed et al., 2025). Ko & Geiping (2026) track representation-space trajectories of multi-turn conversations with an interlocutor. This protocol has one model and no second policy. Wang et al. (2026) keep the prompt in cache and evict generated reasoning tokens. We evict the seed and continue the tail. StreamingLLM (Xiao et al., 2024) studies attention-sink retention and finite-cache streaming, including a sliding-window recomputation baseline. Its long-stream evaluation concerns perplexity on an external text corpus. Here the evolving state consists of the model's own generated text, the seed is fully evicted, and the outcomes are completion, repetition, and seed-conditioned separation. The contribution is this empirical combination; a priority claim for streaming or multi-million-token operation is not made.

Chen et al. (2026) separate the horizon of a task, the number of tokens a model can attend, and memory as a system that persists across steps. We use that separation as terminology. W is the imposed window. Turnover T/W is how long the process was observed. We install no retrieval, scratchpad, or summary. The paper does not show that long-horizon agents thereby gain a memory, and it does not offer a remedy for drift, task failure, or long-text quality. Training-time collapse on recursively generated data (Shumailov et al., 2024) is a different object: these runs never retrain.

The contribution is the combination of five choices. The original prompt is accounted for until it has fully left W. Completion, textual lock, and seed discriminability are measured separately, so a lock among survivors is not mistaken for a high probability of reaching the horizon. Family and post-training comparisons keep the cells that fail. Selected locks are observed far past the horizon with the censoring time written down. The interpretation distinguishes these observations from recovered facts, a semantic half-life, validated states, or permanent regimes. A half-life, a current between semantic states, and a validated Markov state model are not part of that contribution. The molecular-dynamics toolkit used in an early stage (Wu & Noé, 2020; Paul et al., 2019) is recorded in Section 4.5 because the required validation failed.

3 Methods

3.1 Block re-prompt after attempts to finish

Protocol P1 sends Tail_W as a new prompt at every step. It is the only protocol in the results below. A native sliding mask with a continuous key–value cache, protocol P2, was not measured. During one call the model can attend to tokens it has just produced inside that same block, up to B tokens, so P1 is also not a token-by-token slide. Conclusions apply to this self-continuation. They do not automatically describe ordinary chat, tool-using agents, or a cache that is only trimmed at the left edge.

The harness does not treat end-of-sequence as the end of a trajectory. On the occupancy panel, domain trajectories in the last quarter of the run have mean block fill 0.853 with interval $[0.756, 0.936]$ and a mean fraction of calls ending other than by the length limit of 0.359. Generation continues because the stop is overridden. No provider-side minimum-token floor is applied.

The analysis unit is a chunk of 1024 generator tokens, non-overlapping, cut from the generated stream only. The seed is not part of a chunk. Turnover is T/W. A last-band contrast at turnover band 12 is a measurement after the horizon, which is turnover 1. It is not evidence that the lock began after the horizon.

3.2 Two designs, one kernel

Both designs use Equation (1) with W=4096 and B=1024, so B/W=0.25, at temperature 0.3. They are not pooled.

The occupancy panel uses the hosted instruct generator `or-qwen3-8b`, continuation mechanism `raw_completion`, served provider Alibaba on completed steps, and a target of twelve turnovers. Ten domain texts are each run twice. Twin texts are a separate, underpowered contrast and are not a headline. The panel was generated twice. The second generate is the replication used as evidence in Section 4.2. The first generate's embedding directories are not recoverable (Section 3.4).

The family matrix is local. Base and post-trained checkpoints receive the same seed bytes (`raw_bytes`). The primary quantisation is NF4. INT8 rows are limited concordance controls. The prescribed horizon is twelve turnovers, T=49152 tokens. Families are Qwen3-8B Base and Instruct, OLMo-3-7B along Base, SFT, DPO, and RLVR, Ministral-3-8B Base and Instruct, and Gemma-4-12B Base and the instruction-tuned checkpoint. Each NF4 cell contains ten fixed seed texts with four stochastic descendants per text, giving forty planned trajectories. A trajectory that terminates after repeated empty completions stays in the completion count. It is a failure to reach the horizon, not an unlocked success. A later cohort continues eight already-locked cells far beyond twelve turnovers (Section 4.4).

The seed bank has one continuation-shaped text per domain label. Topic, lexicon, and register travel together. A gap between seeds is dependence on that initial text. It does not separate meaning from style.

3.3 Completion, lock, gap, and escape

Four quantities are reported separately.

Completion is the number of planned trajectories whose status is completed at the prescribed token horizon. It is not the lock rate among those that finished.

A textual repetition lock is a surface verdict. A chunk is looping when its 3-gram repetition rate exceeds 0.083, the 99th percentile of a Carroll-and-Darwin reference at the same chunk size. A trajectory is degenerate when at least half of its post-horizon chunks are looping, or when the median late-window shingle Jaccard reaches 0.0122. These are alternative criteria. On the family matrix a lock is confirmed after three consecutive chunk rows have the per-chunk looping flag. Three chunks of 1024 tokens are 0.75 turnover, which is the earliest time the rule can fire. The rule reports a near-recurrent textual regime. It is not an exact period-1 test, and it is not a semantic state. Only a validated Markov macrostate would earn that name. None did (Section 4.5).

The seed-conditioned gap at a turnover band is $G = D_{\text{between}} - D_{\text{within}}$, with cosine distance on L2-normalised chunk embeddings. D_{within} averages pairs that share an initial text. D_{between} averages pairs that do not. A sign is established when the published 95% interval excludes 0. An interval that includes 0 means the sign is not established. An unidentified band, with no between-seed pair, is not a sign flip and is not drawn as zero. The occupancy replication uses a multiplicity-preserving trajectory bootstrap. The family matrix uses a seed-cluster bootstrap that resamples initial texts and preserves the multiplicity of their descendants. Chunks are not treated as independent replicates. Leave-one-seed-out analyses accompany the eligible cells; sparse cells that do not support this analysis remain labelled as such. The two intervals are not the same estimator and are not pooled. Representation spaces are BGE-M3 and Qwen3-Embedding-8B on the family matrix, and those two plus Gemini-Embedding-001 on the occupancy panel. Spaces are not placed on one scale.

Escape requires three consecutive non-looping chunk rows after a confirmed lock. On the long cohort the clock starts at the first token of the twelve-turnover series. τ_{escape} is blank when no escape is confirmed. The censoring time is the common stop, not an estimated mean.

3.4 What can be recomputed

Named run directories for the first occupancy generate and its embeddings are gone. The committed last-band intervals from that panel are archival analysis outputs. They cannot be recomputed from the original vectors, and a new draw does not restore them. The replication is a new 28-trajectory generate. It is the primary evidence for the occupancy gap. The archival intervals remain in Section 4.2 as a historical record of that one panel. Loss of those directories does not license a claim that every earlier number in the project is unreproducible.

4 Results

4.1 Repetition, often confirmed before the horizon

On the occupancy panel, 19/20 domain trajectories meet the lock rule, and 10/10 domain seeds have at least one lock. The trajectory that does not lock is one of the two `love` runs (1/2). A Clopper–Pearson interval for 19/20 is $[0.751, 0.999]$. This is one instruct generator, one protocol, and temperature 0.3. It is not a frequency for language models as a class. On a separate hosted axis at the same W, with n=4 per cell, `or-gemma-4-31b` is 0/4 at both temperature 0.3 and temperature 1.0, while `or-gpt-oss-120b` is 4/4 at both.

The family-matrix lock table for Qwen3-8B shows when the flag fires. After collapsing the two embedding copies of each trajectory, all 86 completed trajectories are confirmed locks and none has a confirmed escape through twelve turnovers. Of the 69 NF4 trajectories, 67 are confirmed at 0.75 turnover (Base 31/31, Instruct 36/38) and the other two Instruct trajectories at 1.00. Full eviction is turnover 1. The value 0.75 is also the floor of the confirmation rule. The table therefore records repetition that the rule can mark before the seed has fully left the window, or, for two trajectories, as that exit occurs. It does not record a regime that is first detected only after the horizon. The same file's smaller INT8 rows are 16/17 at 0.75 and 1/17 at 1.00, which is the same pair of times and not a separate onset result.

4.2 Dependence on the initial text

The primary evidence is the occupancy replication. The design has ten fixed domain texts and twenty planned domain trajectories. Last-band gaps are positive in all three representation spaces (Table 1). In BGE-M3 the gap is 0.208 with interval $[0.091, 0.314]$. Qwen3-Embedding-8B is 0.400 $[0.221, 0.590]$. Gemini-Embedding-001 is 0.152 $[0.073, 0.239]$. Each interval excludes 0 from above. Levels are not interchangeable across spaces. Four domain trajectories stop at 47 chunks rather than 48, so the last band has n_{within}=6 rather than a full within-seed pair set. Band 10 still excludes 0 on the full pair set. It is not the pre-registered last-band number.

This is dependence on the initial condition after that text has left the window. Same-seed trajectories stay closer than different-seed trajectories. The gap does not show that the model remembers original facts, can recover the prompt, or preserves meaning apart from style. Related trajectories may simply continue similar

Table 1: Primary occupancy evidence: last-band seed-conditioned gap on the replication generate, ten domain texts, trajectory-bootstrap 95% interval. Separated means the interval excludes 0 from above. Last-band n_{within}=6. Generate `s5-lock-occupancy-repl-20260907T091450Z-e8452acb`.

Space	gap	95% interval	separated
BGE-M3	0.208	$[0.091, 0.314]$	yes
Qwen3-Embedding-8B	0.400	$[0.221, 0.590]$	yes
Gemini-Embedding-001	0.152	$[0.073, 0.239]$	yes

Table 2: Historical occupancy panel. Published last-band intervals from the first generate. Source vectors are not recoverable (ADR-0021). Not the primary evidence; do not substitute these intervals for Table 1. n_{within}=9.

Space	gap	95% interval	separated
BGE-M3	0.201	$[0.065, 0.332]$	yes
Qwen3-Embedding-8B	0.390	$[0.151, 0.593]$	yes
Gemini-Embedding-001	0.150	$[0.029, 0.266]$	yes

families of repetitions. On this replication the degeneracy flag is true for 20/20 domain trajectories, so the gap is a gap among locked runs.

The first occupancy panel is historical (Table 2). Its published intervals also exclude 0 in the same three spaces, and a recomputation from the committed seed-pair matrix matches the published points to within 0.002. Leave-one-seed-out gaps on that matrix stay positive; the smallest is 0.128. Those checks use the committed matrix. They do not restore the trajectory-bootstrap intervals, because the embeddings are gone. Table 2 is not the primary evidence, and its intervals are not the replication's intervals.

4.3 Families and checkpoints

Completion at 49152 tokens is a different fact from the lock among runs that get there (Table 3). The rows are not a pooled frequency and not one effect of instruction tuning. Identical or neighbouring completion counts are not a demonstrated equivalence.

Qwen3-8B Instruct reaches the horizon more often than Base (38/40 against 31/40). The nine Base failures and two Instruct failures are empty completions, kept in the sample. Which seeds survive also changes. Base `recipe` contributes four empty-completion failures and no lock-table row; Instruct `recipe` is 4/4 locked. On the completers, the last-band interval excludes 0 in both spaces, and the published lock table records no confirmed escape. Base BGE-M3 is 0.147591 $[0.0691844, 0.158049]$; Instruct BGE-M3 is 0.151367 $[0.0871001, 0.159001]$. Base Qwen3-Embedding-8B is 0.233299 $[0.117225, 0.244196]$; Instruct Qwen3-Embedding-8B is 0.304616 $[0.191654, 0.315201]$. Agreement of those signs is a statement about completers. It is not an equivalence of the two checkpoints, and it sits next to different completion counts.

Along OLMo-3-7B the number of completed trajectories falls along the post-training line: 27, 19, 3, and 4 of 40. At DPO and RLVR almost all planned runs fail before the horizon. Base intervals exclude 0 in both spaces: BGE-M3 0.0943909 $[0.0287692, 0.107647]$, Qwen3-Embedding-8B 0.189111 $[0.0861319, 0.221421]$. SFT is 0.10178 $[-0.0069993, 0.121757]$ in BGE-M3, so that sign is not established, and 0.1757 $[0.0342047, 0.205525]$ in Qwen3-Embedding-8B, where the interval excludes 0. DPO has three within-seed pairs and no between-seed pair, so there is no last-band gap in either space. That unidentified band is not a negative gap. RLVR is 0.201503 $[-0.134183, 0.201503]$ and 0.334629 $[-0.147232, 0.334629]$. Both intervals include 0, so a positive gap is not established. The published table records no confirmed escape on these OLMo rows. With three or four completers, that escape count is not a rate to set beside Qwen.

Ministral-3-8B Base and Instruct both complete 40/40. This sample shows no completion difference. That equality is not a general null effect of instruction tuning, and it is not pooled with Qwen or OLMo. Last-band

Table 3: Completed trajectories at the twelve-turnover horizon, NF4, temperature 0.3, W=4096, forty planned per cell. Empty completions remain in the denominator.

Family	Checkpoint	Completed
Qwen3-8B	Base	31/40
Qwen3-8B	Instruct	38/40
OLMo-3-7B	Base	27/40
OLMo-3-7B	SFT	19/40
OLMo-3-7B	DPO	3/40
OLMo-3-7B	RLVR	4/40
Ministral-3-8B	Base	40/40
Ministral-3-8B	Instruct	40/40
Gemma-4-12B	Base	38/40
Gemma-4-12B	instruction-tuned	0/40

intervals exclude 0 in both spaces. Base BGE-M3 is 0.138297 $[0.0633819, 0.158376]$ with no confirmed escape in the published table; Instruct BGE-M3 is 0.150187 $[0.0840649, 0.158143]$ with four confirmed escapes. Base Qwen3-Embedding-8B is 0.239919 $[0.126587, 0.271944]$; Instruct Qwen3-Embedding-8B is 0.224079 $[0.139027, 0.228082]$. A positive gap on this family coexists with a non-zero escape count on Instruct.

Gemma-4-12B is the boundary of the positive gap. Base completes 38/40. Base BGE-M3 is -0.010628 $[-0.0377395, -0.00608505]$. The interval lies entirely below 0: an established negative gap. Base Qwen3-Embedding-8B is 0.0141659 $[-0.0208935, 0.0186512]$. That interval includes 0, so the sign is not established. A common cross-space sign is not established. The published Base row records 15 confirmed escapes among the lock-table count that accompanies those 38 completers. The instruction-tuned checkpoint completes 0/40. It has no last-band gap in either space. Absence of a point is not a sign flip. Nothing in the Gemma contrast identifies a causal role for attention layout. Figure 1 collects the NF4 intervals. Cells with no last-band point are omitted there, not drawn at zero.

4.4 Selected locks observed for a long time

Eight continuations were taken from conditions that had already locked: Ministral-3-8B Base NF4 and Qwen3-8B Instruct NF4, seeds `physics` and `love`, descendants `s1` and `s2`. The pre-specified selection rule ranked checkpoints by the number of trajectories still locked at the short horizon, then selected seeds and descendants by lock time and fixed tie-breaks. It therefore targets conditions that already showed persistent repetition, rather than a representative sample of initial texts. It is not a sample of language models, and the two families are counted rather than pooled into a rate. Each trajectory reaches 4,999,168 original tokens, counted from the first token of the twelve-turnover series (L0). The lock mark on every row is 0.75 turnover. There are 0/8 confirmed escapes through this horizon. τ_{escape} is blank, and the censor flag is true. The common stop is 1220.5 turnover, which is 1219.75 turnovers after the lock mark. Figure 2 shows that censoring time.

The observation is strong for these eight locks: the surface regime that was already in place at 0.75 turnover is still in place at the censoring time. It does not estimate how often such a lock arises, it does not identify a mean time to escape, and it does not show that the regime is permanent. A blank escape time under right-censoring is the absence of an observed exit, not a proof of an absorbing state.

4.5 What was not established

The programme also asked for a validated state model, a semantic half-life, a temperature boundary, and a dependence on W that would deserve to be called a scaling law. Those results were not obtained. Failure to obtain them is not evidence that the phenomena are absent in every regime. It is a failure of these measurements, and the paper does not promote them to findings.

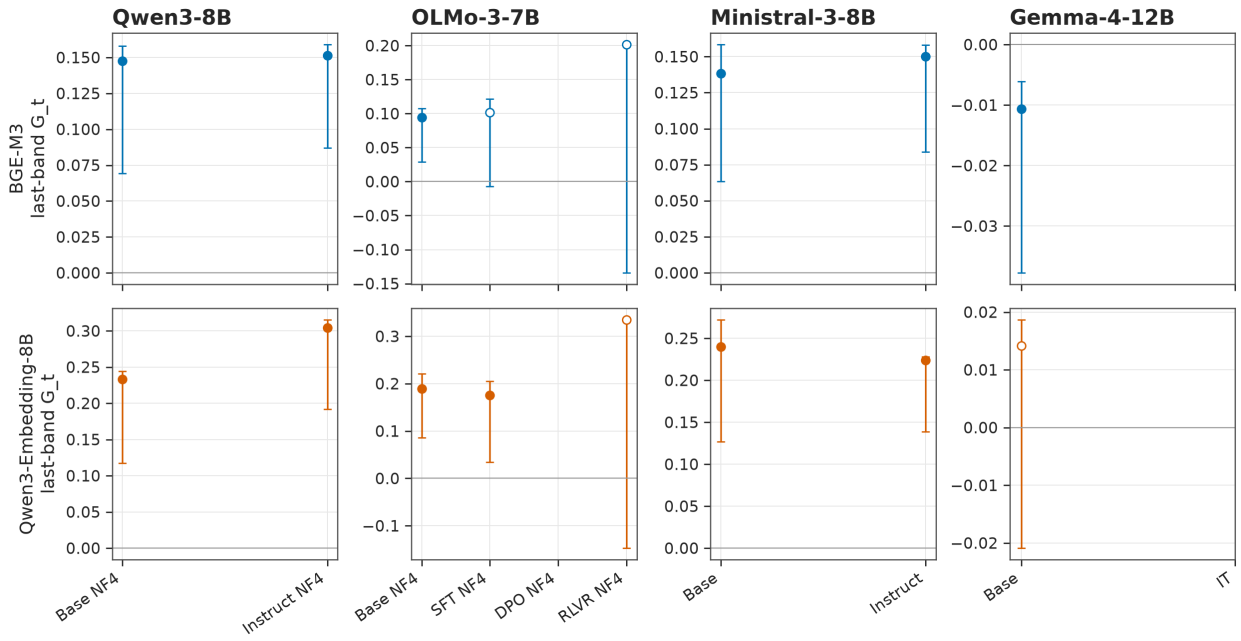


Figure 1: NF4 last-band seed-conditioned gap in the original embedding space. Rows are BGE-M3 and Qwen3-Embedding-8B and are not on one scale. Columns are families and are not averaged. Filled markers: the published interval excludes 0. Open markers: the interval includes 0, so the sign is not established. OLMo DPO and Gemma instruction-tuned have no last-band point and are omitted. Assembly `s13-paperb-synthesis-20261107T085555Z-c0f6b853`.

The state model failed its validation. On every row of the Stage 3 stability table, `validated` is 0. Implied macrostate counts are unstable across the number of microstates and across embedding spaces. A k-means cell in that analysis is a microstate. It is not a semantic state. Hypothesis H1, metastable semantic states as a Markov state model, is unsupported on that sample. A textual lock is not a substitute for the missing validation.

A semantic-memory half-life, in the sense of a time after which half of a defined seed-information measure is gone, was not measured. The between-seed gap is a different quantity. It can stay positive because two trajectories keep repeating similar text. That is not a half-life.

Subdiffusion was not shown to be semantic retention. On the Stage 4 temperature–window grid for `or-qwen3-8b`, a clean mean-squared-displacement exponent is defined only at temperature 1.5, where the non-looping cells live, and every defined exponent is below 1. At temperatures 0.3, 0.7, and 1.0, both W=4096 and W=8192 are 4/4 looping (n=4 per cell), so there is no clean exponent to interpret. At temperature 1.5, W=4096 is 0/4 looping and W=8192 is 2/4. Repetition itself pins a trajectory to a small region of representation space. A confined exponent on a looping panel, or a subdiffusive exponent on the one non-looping temperature, is not a proof that meaning was retained. The 4/4 against 0/4 contrast at W=4096 is not a temperature law for semantic regimes. Two windows on one generator are not a scaling law in W.

Currents between semantic states were not obtained. There is no validated macrostate on which to put a current. The counterfactual twin-seed contrasts, reactor and waterloo, have only two matched stochastic pairs per twin family and are underpowered. Their last-band rows are not reported here as a divergence result, and an equivalence of the twin continuations is not claimed.

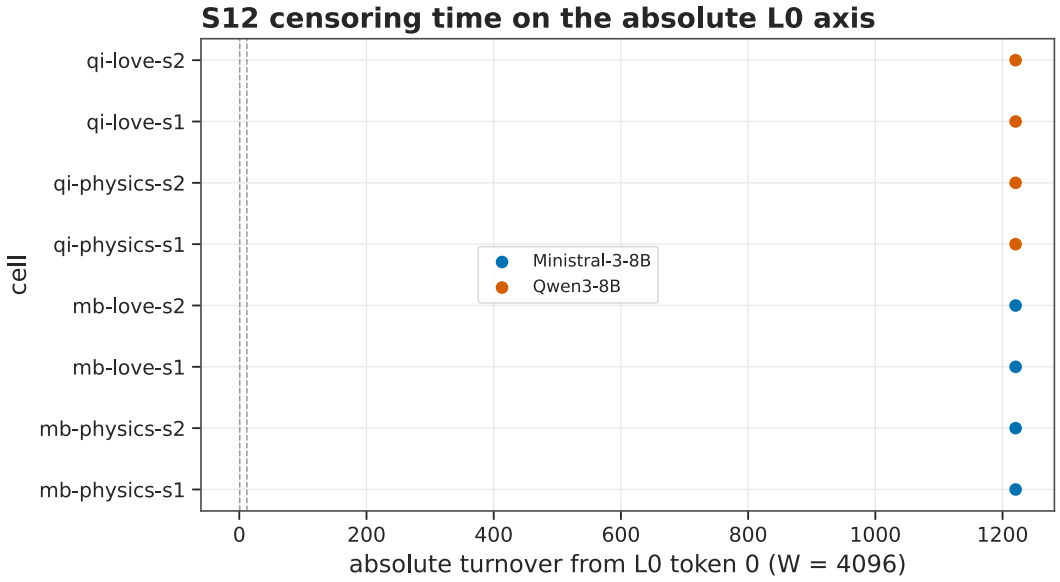


Figure 2: Censoring time for eight selected locks, on the clock that starts at the first token of the twelve-turnover series. All eight cells sit at turnover 1220.5. Dashed lines mark the lock mark at 0.75 and the end of that series at 12.0. No confirmed escape was observed through the common stop. This is not an estimated mean escape time.

5 Limitations

The process is a re-prompt. Each block was produced from a fresh prompt containing the current tail. The harness repeats this call after a full block as well as after an early stop. A continuous key–value cache under a sliding mask was not measured. The results therefore leave the effect of a continuous cache unresolved. The same restriction keeps the results from being a description of ordinary chat or of an agent loop.

Topic and style are not separated. Each domain label is one text. Content, lexicon, and register are entangled. The seed-conditioned gap is therefore a weak instrument for any claim about semantic memory. Related locks can share a style of repetition and produce a positive gap with no preserved proposition.

The family matrix is a narrow regime. Temperature is 0.3, the window is one value of W, the serialization is raw bytes, and the headline quantisation is NF4. INT8 controls plan ten trajectories per cell, not forty. Some controls have only three or four within-seed pairs, and their intervals include 0. They do not isolate the effect of quantisation; no matched BF16 control was run. BGE-M3 was computed locally. Qwen3-Embedding-8B for the family matrix was a hosted endpoint, so the two spaces are also two pipelines. Provider non-determinism was not re-measured on these horizons. On an earlier hosted audit, exact-match rates were 100% for glimmer, 60% for Qwen, 20% for Gemma, and 20% for both GPT-oss checkpoints.

Conditioning on completion changes the sample. OLMo DPO contributes three completers and no between-seed pair. Gemma instruction-tuned contributes none. On Qwen3-8B the Base lock table is missing `recipe`, which Instruct retains at 4/4. A gap estimated on survivors answers a different question from a gap estimated on the planned design. The paper reports both the completion count and the conditional gap, and does not treat the second as if the first had been balanced.

The first occupancy embeddings are gone. This limitation is local to that panel. The replication strengthens the sign of the occupancy gap, and Table 1 can be recomputed from its own run directories. The archival intervals in Table 2 cannot. The paper does not promise that every earlier number, outside this panel, can be regenerated bit for bit. Hosted sampling was already not deterministic.

The long cohort was selected, and every horizon here is finite. The eight long trajectories were chosen from locks that had already persisted, by a rule fixed before the extension. Twelve turnovers is the family-matrix horizon. 1220.5 turnovers is a censoring time. Neither is an infinite afterlife. The occupancy replication's last band is twelve turnovers as well.

No remedy is tested. The measurements do not show how to improve agent memory, task completion, or the quality of long text. Nucleus sampling, repetition penalties, and retrieval were not the interventions. A diagnostic description of a failure mode is not a fix.

6 Discussion

After the original prompt has been physically removed, self-continuation under this block re-prompt can preserve textual repetitions for a long time, and it can preserve a statistical dependence on the initial text. Completion, and the stability of both the lock and the dependence, differ across the families and checkpoints that were run. These are distinct outcomes of the measured kernel.

The positive gap is established on the replication, on Qwen3-8B and Ministral-3-8B NF4 in both spaces, and on OLMo-3-7B Base in both spaces. It is mixed on OLMo SFT, unidentified on OLMo DPO, and not established on OLMo RLVR. On Gemma-4-12B Base it is an established negative gap in BGE-M3 and an unestablished sign in Qwen3-Embedding-8B. Positive discriminability has a boundary inside this study. The boundary is a limit on generalisation. It is not an architectural explanation.

On the Qwen3-8B lock table, the repetitions that carry the late gap are usually confirmed before the seed has left the window. The long cohort then shows that eight such locks, selected from conditions with many persistent locks, have no confirmed escape through 4,999,168 original tokens each. Longevity of a selected lock is a different claim from a late-onset regime, and a different claim from a universal half-life. The measurements establish neither semantic memory, a validated set of metastable states, nor circulation among such states. The state model failed validation. The half-life was not measured. Currents between semantic states were not obtained. The temperature grid did not establish a boundary between semantic regimes, and the two windows did not establish a scaling law.

The literature already contains repetition (Holtzman et al., 2020; Xu et al., 2022), a Markov description of finite-window generation (Zekri et al., 2024), paraphrase cycles (Wang et al., 2025), and semantic drift under whole-text iteration (Geng et al., 2026). What this record adds is the process distinction and the bookkeeping: eviction of the original prompt is explicit, completion is not hidden inside a conditional lock rate, negative cells stay in the tables, the long observation is censored, and the interpretations that the geometry tempts are checked against those facts. Zekri et al. (2024) provide a uniqueness result for the full-support token-sampling kernel under their assumptions. A positive gap in an embedding of locked text is compatible with more than one long-run region of that embedding. It is not a second stationary distribution of the token chain.

The practical reach of the result is diagnostic. Someone running a long self-continuation can use completion, lock, and seed gap as separate symptoms, and can expect those symptoms to depend on the checkpoint. The paper does not say which change of decoding, quantisation, or memory harness would help. That would be a different experiment.

7 Reproducibility

This manuscript assembles closed generation and analysis runs. It adds no generated trajectories, embeddings, or re-estimated results. Experiment configurations and manifests record the code revision, random seeds, tokenizer revisions, provider, and checkpoint settings. The local checkpoint pins and quantisation recipe are in `configs/models/generators_local_paperb.yaml`; the initial text bytes are in `configs/seeds/seed_bank_v1.yaml`. Each included figure retains its committed tidy data and metadata alongside the plotted image. The assembly gate `afterlife review -stage s13` checks the mechanical integrity of the synthesis record. It does not establish scientific validity or restore the missing archival vectors.

Identifiers for the claims above:

- Occupancy lock table: `s5-degeneracy-20260906T030145Z-deb4c3bd` (generate `s5-lock-occupancy-20260905T164327Z-6780902f`, directories not recoverable).
- Occupancy replication generate: `s5-lock-occupancy-repl-20260907T091450Z-e8452acb`; embeddings `s5-embed-lock-occupancy-repl-20260907T175131Z-4e48f831`, `s6-embed-third-space-20260907T181515Z-bcde0a11`.
- Qwen3-8B NF4: `s9-paperb-qwen-base-nf4-20260912T014645Z-f55767bc`, `s9-paperb-qwen-instruct-nf4-20260911T054758Z-c54e2ab6`.
- OLMo-3-7B NF4: `s10-paperb-olmo-base-nf4-20260916T204957Z-6d7b4be3`, `s10-paperb-olmo-sft-nf4-20260917T142201Z-2b7b5b6e`, `s10-paperb-olmo-dpo-nf4-20260918T000412Z-64ad9f9f`, `s10-paperb-olmo-rlvr-nf4-20260918T022536Z-30b5b3fc`.
- Ministral-3-8B and Gemma-4-12B: `s11-paperb-ministral-base-nf4-20260919T130616Z-5896956b`, `s11-paperb-ministral-instruct-nf4-20260920T065557Z-a3093ea2`, `s11-paperb-gemma4-base-nf4-20260921T003925Z-6cae9981`, `s11-paperb-gemma4-it-nf4-20260922T160427Z-3b5b9427`.
- Published family-matrix gaps: `s13-paperb-synthesis-20261107T085555Z-c0f6b853`.
- Eight long trajectories: the `run_id` list in `artifacts/stage-12/tau_escape/tau_escape.meta.json`.
- State-model validation: `s3-dynamics-20260901T204521Z-f21d5908`, `s3-dynamics-20260901T204549Z-b7c4d0c9`.
- Temperature grid: `s4-w4096-new-temps-20260904T103121Z-589c8eb1`, `s4-w8192-20260904T120057Z-ce82ce55`.

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